# Nathan McMillan

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## SUMMARY

Computer Science student at Clemson University with a minor in Artificial Intelligence, graduating December 2026. Experienced in software development, machine learning, computer vision, and robotics.
Seeking entry-level roles in **AI/ML Engineering, Software Engineering, Computer Vision, or Applied AI**.

## EDUCATION

**Clemson University** **August 2022 – December 2026**
*B.S. in Computer Science, Minor in Artificial Intelligence* *Clemson, SC*
*GPA: 3.2/4.0*
**Clubs & Organizations:** Clemson Forge, Member (2024)

## PROJECTS

### VLM for Wildfire Detection

**Feb 2026 – April 2026**

- Reproduced a teacher–student smoke-segmentation pipeline, using SAM to generate masks and PIDNet-S/M as lightweight student models.
- Qwen2.5-VL sometimes rejected real faint-smoke masks when confidence thresholds were too strict; the revised approach preserved uncertain teacher labels.
- Matched reported baseline mIoU: PIDNet-S 0.591 vs. 0.594; PIDNet-M 0.605 vs. 0.606. The VLM-filter and fused variants reached 0.231 and 0.161 mIoU, respectively.

### Vision-Language-Action Robot Manipulation

**September 2026 – Present**

- Developing a simulation-based robot-manipulation study using LIBERO, MuJoCo, and robosuite, comparing vision-only, image-language, and Grid Sampler policies across two tasks
- Planning 60-100 simulated demonstrations and held-out evaluations of task success, inference latency, and visual-token usage

### Embodied AI for Automotive Manufacturing Inspection

**August 2026 – Present**

- Developing a CU-ICAR vehicle-inspection prototype and contributing to an NSF STRIDE proposal focused on adaptable edge AI and traceable inspection data
- Curated 500+ images with assisted labeling; established an initial YOLO26 demo baseline of 63% mAP@0.5 across three component classes and demonstrated 10-minute local OpenVINO inference.

### Ambient Health Feedback App

**Feb 2026 – April 2026**

- Built a SwiftUI app that translated Apple Health data into ambient feedback about physical well-being
- Conducted heuristic-guided usability testing to evaluate feedback clarity and perceived interruption.

### Good Driver Incentive Program

**Jan 2026 – April 2026**

- Built a React frontend and Python/MySQL backend on AWS; added CI/CD and automated API tests.

### TigerChow Web App

**Feb 2025 – Apr 2025**

- Built an on-campus ordering system using AWS Lambda, API Gateway, and DynamoDB

## RELEVANT EXPERIENCE

**Student Intern - CCIT** **Clemson, SC | May 2025 – July 2026**
- Supported the installation, maintenance, and troubleshooting of classroom systems, ensuring reliable operation for daily academic use

## TECHNICAL SKILLS

**Languages**: Python, Java, C++, C, JavaScript, SQL, HTML/CSS, Swift
**Frameworks & Tools:** React-Native, SwiftUI, PyTorch, TensorFlow, Scikit-learn, Git, GitHub, JUnit, Postman, CI/CD, Unreal Engine 5
**Cloud & Technologies**: AWS, Azure, REST APIs, Linux, Raspberry Pi, HealthKit, Computer Vision, IoT